



**UNIVERSITY
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Introduction to Data Science (DATA11001)

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D.R.A.M.A.

Project's technical report

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Abstract

This project applies data science and machine learning techniques to predict movie performance and understand audience sentiment. Using data from IMDB, TMDB, and HuggingFace, it combines movie metadata with textual reviews to extract meaningful insights. Linear Regression and Random Forest models were trained to estimate ratings and revenues, while clustering revealed audience preference patterns. All results were integrated into a web application that allows users to explore key statistics, visualize sentiment trends, and predict film outcomes, supporting data-driven creative and business decisions.

Table of contents

Introduction.....	2
Methodology.....	2
Data collection.....	2
Data processing.....	2
Exploratory data analysis.....	3
Machine learning models.....	4
Communication of results.....	4
Evaluation of results.....	5
Conclusions.....	5

Introduction

This project applies data science and machine learning techniques to understand and predict audience sentiment toward films. The goal is to transform unstructured text from movie reviews into structured, actionable insights that can guide creative and business decisions. Using a dataset that combines movie metadata (such as genre, director, and cast) with textual reviews, the system extracts key features, classifies sentiment, and identifies underlying themes. By integrating natural language processing methods, including keyword extraction, sentiment analysis, and clustering, the project uncovers meaningful relationships between film attributes and audience perception.

From a technical perspective, multiple predictive and analytical models were implemented to explore different aspects of movie success. Linear regression and decision tree models were used to estimate audience ratings and profitability, while unsupervised clustering methods such as K-means and t-SNE visualization revealed latent audience preference patterns. These models together form D.R.A.M.A. — the Data-Driven Revenue & Audience Metrics Analyst — a system that enables film professionals and analysts to compare films objectively, detect trends, and align production strategies with audience expectations.

Methodology

Data collection

Three main datasets were utilized in this project, each serving a distinct analytical purpose. The first dataset comes from IMDB (Internet Movie Database), containing approximately 85,000 movies with detailed metadata such as title, release year, runtime, genres, directors, writers, production companies, and user ratings. This dataset provided the foundation for the project's predictive modeling component, as it included numeric and categorical variables suitable for regression and tree-based analysis. The IMDB data helped quantify relationships between production attributes and audience ratings.

The second dataset was obtained from TMDb (The Movie Database), accessed through its open API. It contained around 55,000 films, including metadata such as popularity scores, vote averages, number of votes, language, and revenue. This integration allowed the project to model both critical reception and commercial performance across diverse movie genres.

The third dataset was sourced from Hugging Face, under the repository `Daksh0505/IMDB-Reviews`, which contains approximately 50,000 audience reviews evenly distributed between positive and negative sentiment labels. This dataset formed the basis for the project's text analysis and clustering components.

Data processing

For the IMDB and TMDb datasets, the preprocessing phase focused on cleaning, normalizing, and integrating the structured metadata. Missing values in attributes such as runtime, revenue, and vote count were imputed using mean or median values depending on the data distribution. Categorical variables (e.g., genres, production companies, and countries) were standardized by unifying inconsistent labels and encoding them into a machine-readable format using one-hot and label encoding techniques. Outliers in numerical variables, particularly in budget and revenue, were handled through percentile-based

trimming to reduce skewness. To merge both datasets, the IMDB identifier *imdb_d* was used as the primary key, ensuring accurate alignment between corresponding movie records. For the Hugging Face review dataset (Daksh0505/IMDB-Reviews), preprocessing focused on preparing the unstructured text for NLP and clustering tasks. Each review was cleaned by removing punctuation, HTML tags, and non-alphabetic characters, followed by conversion to lowercase to maintain uniformity. Stopwords were removed to reduce noise, and tokenization was applied to split the text into meaningful units. Stemming was then used to normalize words to their base forms, enhancing consistency in subsequent analysis. Sentiment labels (positive or negative) were encoded as binary numeric values to facilitate supervised learning tasks. The cleaned and tokenized corpus was then vectorized using TF-IDF representations, allowing the system to extract relevant features for sentiment classification and unsupervised clustering of audience opinions.

Exploratory data analysis

For the IMDB dataset, the exploratory analysis concentrated on a comprehensive examination of genre distribution, director performance, and audience voting behavior. Genre frequency statistics revealed that Drama, Comedy, and Romance dominate the dataset, while less common genres such as History and Biography exhibited narrower but more specialized audience ratings. Director performance analysis was conducted by aggregating the average rating per director across all their films, allowing identification of consistently high-performing creators. Patterns in runtime, release year, and rating distribution indicated a tendency for mid-length films and more recent releases to receive a higher number of votes, reflecting increased audience engagement with contemporary content.

For the TMDB dataset, analysis focused on understanding the financial performance and its relation to critical reception. Scatterplots of budget vs. revenue and budget vs. rating demonstrated a positive but nonlinear relationship between investment and commercial success, while the correlation between budget and ratings remained relatively weak. Additional insights were drawn from ranking the films by highest budget, highest revenue, and highest ROI.

For the IMDB Reviews dataset from Hugging Face, exploratory text analysis revealed key sentiment tendencies and frequent expressions among users. Preliminary examination of the textual corpus highlighted recurring sentiment-driven keywords that later informed the NLP preprocessing and feature extraction process.

Across all datasets, feature engineering played a crucial role in transforming raw data into meaningful predictors.

Machine learning models

To predict movie ratings and revenues, multiple machine learning models were implemented and evaluated. The goal was to quantify the relationship between film attributes (such as cast, crew, budget, and audience sentiment) and measurable outcomes of success.

The Linear Regression model served as the baseline due to its interpretability and ability to reveal direct relationships between predictors and the dependent variables. Although its predictive power was moderate, linear regression was valuable for identifying key features

with the strongest linear correlations, particularly those related to budget, genre count, and average sentiment score.

The Random Forest Regressor was implemented to capture nonlinear interactions and higher-order feature dependencies that the linear model could not fully represent. Trained on a similar dataset with 100 decision trees, it achieved improved results. The ensemble structure of Random Forests provided robustness against overfitting and allowed feature importance extraction, which proved instrumental in identifying the most influential factors on both ratings and revenues.

Finally, clustering analysis was conducted to uncover hidden structures and audience preference patterns. Using a combination of numerical and text-derived features (such as sentiment vectors and genre embeddings), unsupervised clustering methods enabled grouping of films with similar audience responses and production profiles. This exploratory segmentation provided qualitative insights into the types of films that tend to perform similarly in both critical and commercial dimensions.

Communication of results

To effectively communicate the analytical outcomes, an interactive web application was developed as the final component of the project. This platform serves as a bridge between data science findings and practical decision-making for the target audience (primarily movie producers, analysts, and creative executives). Within the app, users can explore a wide range of statistical insights obtained during the EDA phase, such as genre distributions, director performance metrics, runtime patterns, and audience voting behaviors. These visualized metrics enable professionals to better understand the underlying factors influencing audience sentiment and film performance.

In addition to descriptive analytics, the web app integrates the predictive models built during the project. Users can input a combination of directors, writers, and lead actors to estimate both the expected rating and revenue of a potential film using the trained Linear Regression model. This predictive feature provides a data-driven perspective for assessing creative choices before production. Beyond numerical prediction, the platform also showcases results from the clustering analysis, including Word Cloud Analysis, Aspect-Based Analysis, and genre-level insights such as Genre vs. Aspect and Aspect Preferences Comparison. These visualizations, along with a t-SNE representation of clustered reviews, allow users to intuitively explore audience preferences and the thematic structure behind film reception.

Evaluation of results

The predictive and exploratory models yielded complementary insights into the factors influencing movie performance. The Linear Regression model provided a transparent baseline for understanding how individual features relate to ratings and revenues. With a Test R^2 Score of 0.2882, RMSE of 1.0625, and MAE of 0.8160, the model demonstrated a moderate ability to capture the variance in audience ratings. Despite its limitations in modeling nonlinear relationships, it proved valuable for feature interpretability.

In contrast, the Random Forest Regressor achieved slightly higher predictive accuracy. This improvement highlights the model's capability to capture complex, nonlinear interactions

between features that linear regression could not. Feature importance analysis revealed that ROI, sentiment polarity, and director consistency were among the most influential predictors.

While regression models focused on quantitative prediction, the clustering analysis provided a complementary qualitative perspective. By grouping films based on sentiment, genre composition, and production characteristics, the clustering revealed latent structures in audience preferences. Each cluster corresponded to distinct audience segments. These groupings supported deeper understanding of why certain movies perform similarly, offering a descriptive dimension that numerical models alone could not provide.

Together, the regression and clustering results establish a comprehensive analytical framework: regression models quantify how much individual factors affect success, while clustering models reveal why similar films resonate with audiences in comparable ways.

Conclusions

Although the Linear Regression model was conceptually straightforward, its implementation required more time and fine-tuning than initially expected. It is important to mention that while the linear regression provided valuable interpretability and transparency, its limited capacity to capture nonlinear relationships between features and outcomes constrained its predictive accuracy. This reinforced the understanding that in real-world data, especially within the entertainment industry, success rarely depends on linear interactions alone.

The clustering analysis introduced additional challenges, especially when dealing with unstructured textual data such as audience reviews. Sentiment analysis added layers of ambiguity, as linguistic nuances and contextual expressions are difficult to fully capture through automated processing. Nevertheless, clustering proved highly valuable for uncovering hidden audience patterns and thematic similarities among films. Beyond this project, such segmentation could be applied to marketing strategies, personalized content recommendation, and script development.

The web application successfully integrated the predictive and exploratory results, providing a practical interface for users to interact with the data. However, there remains room for improvement, particularly in optimizing the user experience, enhancing interactivity, and integrating more advanced models.

Overall, the project demonstrated how combining predictive modeling with unsupervised exploration can transform raw data into meaningful insights for both creative and business decision-making in the film industry.